

Graduate Deep Reinforcement Learning Paper Chains

Comprehensive Project Report

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Course: Deep Reinforcement Learning

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Contents

1	Introduction	2
2	Related Work	2
3	Methods	2
3.1	Codebase Overview	2
3.2	Chain Descriptions	3
4	Experimental Setup	3
5	Experiments and Results	3
5.1	Value-Based Results	4
5.2	Policy Gradient / Trust Region	4
5.3	Actor–Critic / Max-Entropy	5
5.4	Model-Based	5
5.5	Multi-Agent	5
6	Ablations	5
7	Discussion	5
8	Limitations and Ethics	6
9	Reproducibility Checklist	6
10	Conclusion	7
A	Theoretical Notes	9

List of Tables

1	Entry-point commands (laptop-friendly defaults).	3
2	Value-based results (auto-filled).	4
3	Policy-gradient / trust-region results (auto-filled).	4
4	Actor-critic / maximum-entropy results (auto-filled).	5
5	Model-based results (auto-filled).	6
6	Multi-agent results (auto-filled).	7

List of Figures

1	Value-based learning curve generated from suite runs.	4
2	Policy gradient curve generated from suite runs.	5
3	Actor-critic curve generated from suite runs.	6
4	Model-based curve generated from suite runs.	7
5	Multi-agent curve generated from suite runs.	8

1 Introduction

This project reproduces and analyzes five canonical deep RL paper chains spanning value-based, policy-gradient/trust-region, actor-critic with maximum entropy, model-based, and multi-agent learning. The goals are: (i) implement faithful yet compact baselines, (ii) evaluate them on standardized benchmarks, (iii) compare theoretical guarantees with empirical behaviour, and (iv) provide reproducible artifacts suitable for graduate coursework.

2 Related Work

- Tabular Q-learning convergence [13] and stability refinements (Double DQN [11], Dueling DQN [12]).
- Trust-region methods TRPO [8] and PPO [9]; constrained policy optimization [1].
- Actor-critic advances including A3C [6] and maximum-entropy SAC [2].
- Model-based rollouts from Dyna-Q [10] to ensemble TRPO [3].
- Multi-agent formulations: minimax Q [4], centralized training with decentralized execution (MADDPG [5]), and value factorization (QMIX [7]).

3 Methods

3.1 Codebase Overview

All code lives in `clean_grad_rl/`. Entry scripts implement the five chains (see table 1). Shared utilities handle seeding, environment creation, reproducible logging, and evaluation.

Table 1: Entry-point commands (laptop-friendly defaults).

Chain	Command
Single run	<code>python scripts/run_experiment.py --chain value --algo dqn --env CartPole-v1</code>
Full suite	<code>python scripts/run_suite.py --suite configs/suites/fast_5chain.yaml --mode f</code>
Smoke suite	<code>python scripts/run_suite.py --suite configs/suites/fast_5chain.yaml --mode s</code>
Plot build	<code>python scripts/generate_plots.py --suite-dir outputs/suite_reports --runs-r</code>
Table sync	<code>python scripts/build_report_tables.py --aggregate-csv outputs/suite_reports/</code>

3.2 Chain Descriptions

Value-based. DQN with replay and target networks; optional Double, Dueling, and QR-DQN variants.

Policy-gradient / trust-region. REINFORCE baseline; PPO clip surrogate; TRPO with CG and line search; CPO-lite using Lagrangian penalty for safety cost.

Actor-critic / max-entropy. A2C as synchronous A3C stand-in; SAC with automatic entropy tuning.

Model-based. Tabular Dyna-Q with planning updates; lightweight ME-TRPO-style ensemble dynamics with periodic TRPO updates using imagined rollouts.

Multi-agent. QMIX monotonic value factorization and MADDPG centralized critics on PettingZoo MPE simple_spread.

4 Experimental Setup

Hardware: document CPU/GPU, RAM.

Software: Python 3.10+, PyTorch 2.2.2, Gymnasium 0.29.1, PettingZoo MPE (pinned in `requirements.txt`).

Environments: CartPole/LunarLander, MountainCarContinuous, Pendulum, CliffWalking, MPE simple_spread.

Seeding: `--seed` flag; vector environments offset seeds per worker.

Logging: JSON metrics in `outputs/runs/.../metrics.json`, aggregate CSV in `outputs/suite_reports`

5 Experiments and Results

Populate the following tables after running the scripts.

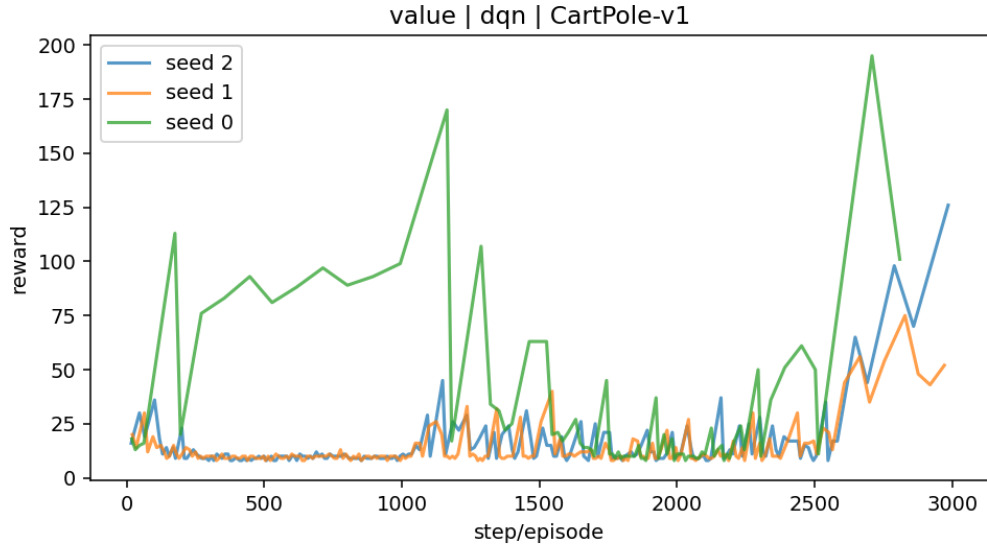


Figure 1: Value-based learning curve generated from suite runs.

Table 2: Value-based results (auto-filled).

Variant	Seeds	Mean Return	Std Across Seeds
DQN	3	206.07	184.89
Rainbow-lite	3	-98.07	117.07

5.1 Value-Based Results

5.2 Policy Gradient / Trust Region

Table 3: Policy-gradient / trust-region results (auto-filled).

Algo	Env	Seeds	Mean Return	Std Across Seeds
REINFORCE	CartPole-v1	3	9.53	0.09
PPO	CartPole-v1	3	213.33	69.85
PPO	MountainCarContinuous-v0	3	-4.94	0.97
TRPO-lite	CartPole-v1	3	252.00	73.24
CPO-lite	CartPole-v1	3	25.33	17.59

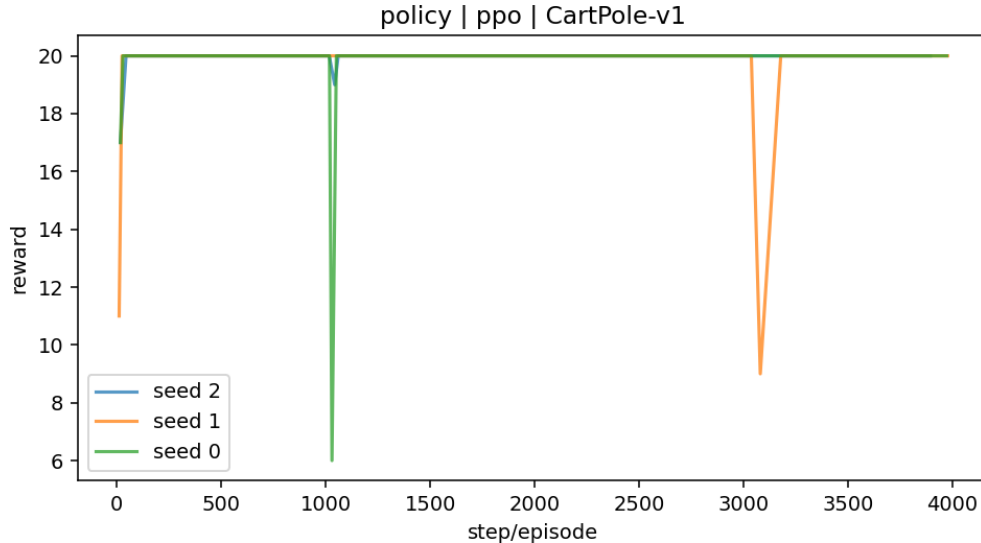


Figure 2: Policy gradient curve generated from suite runs.

Table 4: Actor-critic / maximum-entropy results (auto-filled).

Algo	Seeds	Mean Return	Std Across Seeds
A2C	3	-1654.39	145.76
SAC	3	-890.27	408.71

5.3 Actor-Critic / Max-Entropy

5.4 Model-Based

5.5 Multi-Agent

6 Ablations

Suggested ablations: replay buffer size, Double vs. vanilla DQN; PPO clip range; SAC entropy coefficient fixed vs. auto; Dyna-Q planning steps; QMIX worker counts.

7 Discussion

Relate empirical findings to theoretical expectations (see §4 for setup): overestimation bias reduction, KL constraint stability, entropy-driven exploration, model bias vs. rollout horizon, multi-agent non-stationarity.

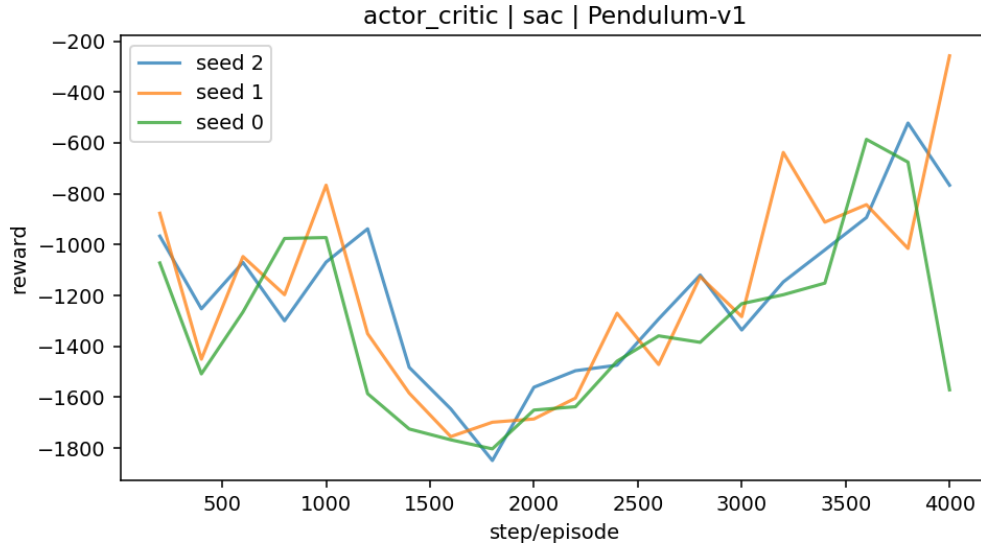


Figure 3: Actor-critic curve generated from suite runs.

Table 5: Model-based results (auto-filled).

Algo	Environment	Seeds	Mean Return
Dyna-Q	CliffWalking-v0	3	-47.52
MBPO-lite	Pendulum-v1	3	-950.00

8 Limitations and Ethics

Note approximations (A2C for A3C, simplified CPO), environment realism limits, and responsible reporting of randomness and hyperparameters.

9 Reproducibility Checklist

- Code and scripts: `clean_grad_rl/scripts/*.py`
- Configs: YAML defaults under `clean_grad_rl/configs`
- Seeds: recorded in logs; set via CLI
- Dependencies: `requirements.txt`
- Hardware/software: state in §4
- Data: environments auto-downloaded when required
- Metrics: JSON/CSV/PNG artifacts in `outputs/runs` and `outputs/suite_reports`

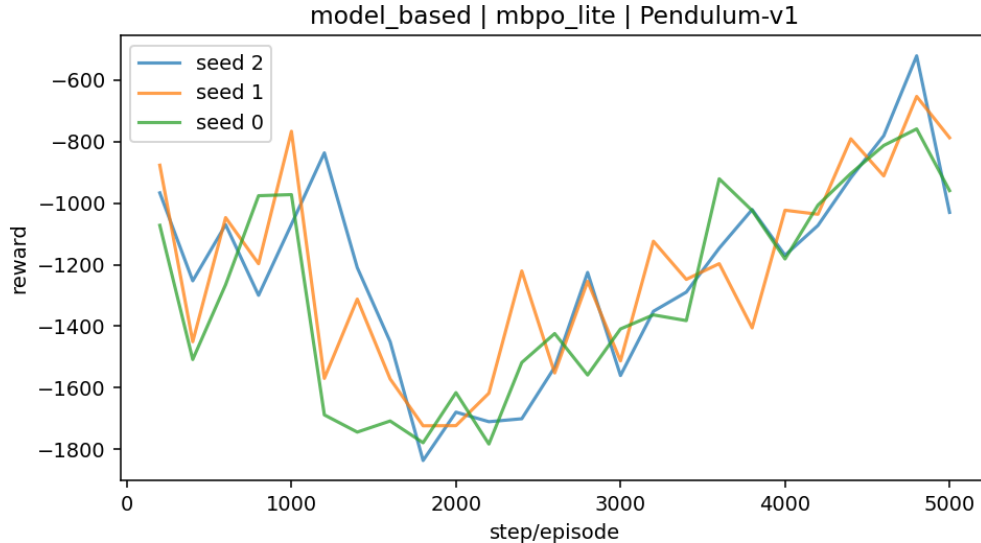


Figure 4: Model-based curve generated from suite runs.

Table 6: Multi-agent results (auto-filled).

Algo	Seeds	Episode Reward Mean
IPPO	3	-72.08
QMIX-lite	3	-34.58

10 Conclusion

Summarize which methods are most stable, sample-efficient, or safe for the tested domains, and outline future extensions (Rainbow, true CPO, PlaNet/MuZero, SMACv2).

References

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- [3] Thanard Kurutach, Aviv Tamar, Ge Yang, Stuart Russell, and Pieter Abbeel. Model-ensemble trust-region policy optimization. In *International Conference on Learning Representations*, 2018.
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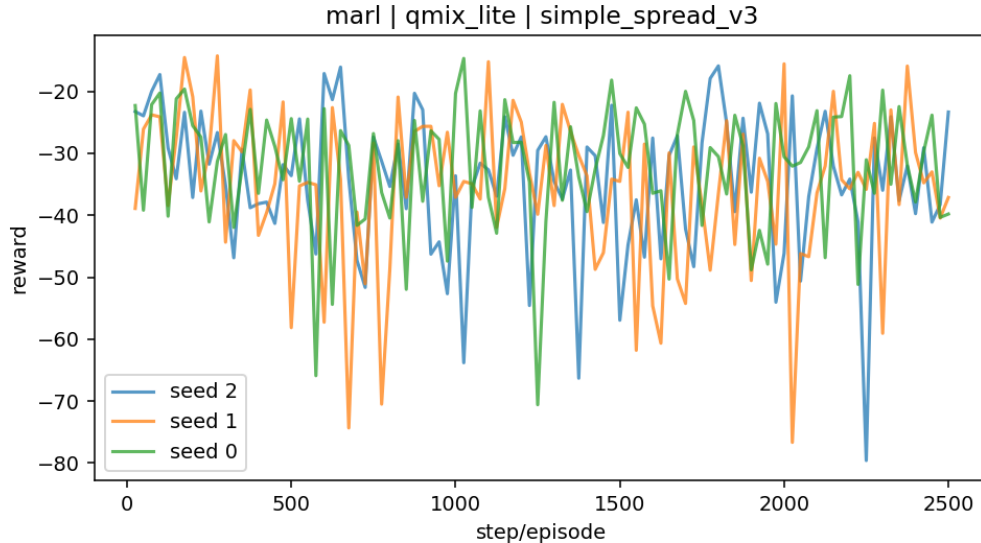


Figure 5: Multi-agent curve generated from suite runs.

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- [13] Christopher JCH Watkins and Peter Dayan. Q-learning. *Machine Learning*, 8:279–292, 1992.

A Theoretical Notes

For detailed derivations and proofs, see `solutions.tex`; reuse its content here as needed.